When does the dominant coefficient of a polynomial continued fraction vanish?

A monotone positivity criterion (machine-checked core),
an unconditional bound for a positive family,
and an explicit vanishing witness

Papanokechi
ORCID: 0009-0000-6192-8273

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Abstract

Let b be a real polynomial of degree d with leading coefficient $\beta_d > 0$, and let q_n be the continuant denominators of the polynomial continued fraction with partial denominators $b(n)$ and unit partial numerators: $q_0 = 1$, $q_1 = b(1)$, $q_n = b(n)q_{n-1} + q_{n-2}$. Writing $P_n = \prod_{k=1}^{n} b(k)$ and $Q_n = q_n/P_n$, the scaled limit $C := R_\infty := \lim_n Q_n$ is the dominant (connection) coefficient of the seed $(1, b(1))$; one has $C \neq 0$ iff q is not a minimal solution. We give a clean account of when $C \neq 0$. First, an elementary *ratio criterion*: once $b > 0$ on a tail and two consecutive q are positive, $\rho_n = q_n/q_{n-1} > b(n) \to +\infty$, so q is dominant and $C \neq 0$. Second, a *positivity-persistence lemma*: if $u_n = 1/(b(n-1)b(n)) > 0$ from some index on and the scaled sequence is then positive, Q_n is nondecreasing and $C \geq Q_{N_0} > 0$; the monotone–bounded core is verified in Lean 4 (Mathlib) with a clean axiom cone (no `sorryAx`). As a corollary, $b(n) > 0$ for all $n \geq 1$ forces $C \geq 1$; in particular the family $b(n) = An^2 + Bn + C_0$ with integers $A, B, C_0 \geq 1$ has $C \geq 1$ unconditionally, which recovers the strict positivity $\delta = \log R_\infty > 0$ used in the companion quadratic growth-law deposit. Third—and this is the point of the falsification-first method—the *universal* statement "$C \neq 0$ for every $\beta_d > 0$" is **false**: we exhibit a real quadratic $b(n) = n^2 - 6.2803109563277\ldots\, n + 8.4205764270426\ldots$ for which $C = 0$ to more than forty digits at 50-digit working precision, with the dominant scale $\prod |b|$ exceeding 10^{43000} at the same index—a genuine minimal seed. The vanishing is produced by the sign-varying transient $(b(2), b(3) < 0)$, not by the tail, and $b(1) > 0$ does *not* imply $C \neq 0$. We are careful to claim only ≥ 1 such zero: the map $(\alpha, \beta) \mapsto R_\infty$ has genuine poles wherever b has an integer root, so a sign-change count is not a zero count. Finally we flag the one genuinely open point: $C = 0$ shows the singular Frobenius coefficient vanishes, but the stronger statement that the associated generating function is *entire* requires a separate unique-nearest-singularity analysis, left as future work.

Epistemic status (SIARC four-class). Each statement below carries one of: **PROVEN** (machine-checked; clean Lean axiom cone $\subseteq$ {`propext`, `Classical.choice`, `Quot.sound`}, no `sorryAx`), **VERIFIED** (high-precision numerics, stated digits above the floor), **STRUCTURAL** (complete elementary hand proof, not yet formalized), **CONJECTURED**, or **FALSE** (disproved by explicit witness). Citations to the literature that were not source-verified for this draft are marked [FLAG(operator): confirm locator]; they are used only for context and no result below depends on them.

1 Setup and the meaning of C

Fix a real polynomial b of degree d with leading coefficient $\beta_d > 0$, and consider the second-order recurrence of continuant denominators

$$q_n = b(n)\, q_{n-1} + q_{n-2}, \qquad n \geq 2, \qquad q_0 = 1,\ q_1 = b(1). \tag{1}$$

The transfer matrix $M_n = \left[\begin{smallmatrix} b(n) & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has $\det M_n = -1$, so solutions of (1) form a 2-dimensional space and the Casoratian (discrete Wronskian) of any two solutions y, z satisfies $W_n = -W_{n-1}$, i.e. $W_n = (-1)^n W_0$. For the numerator/denominator pair this is the identity

$$p_{n+1}q_n - p_n q_{n+1} = (-1)^n, \tag{2}$$

which is **PROVEN** in Lean (§6, `pq_casoratian`).

Assume $b(k) \neq 0$ for all $k \geq 1$ (otherwise the product scaling is undefined; $b(1) = 0$ is excluded outright). Put $P_n = \prod_{k=1}^{n} b(k)$ and the scaled sequence $Q_n = q_n/P_n$. Dividing (1) by P_n gives

$$Q_n = Q_{n-1} + u_n\, Q_{n-2}, \qquad u_n = \frac{1}{b(n-1)\, b(n)}, \qquad Q_0 = Q_1 = 1. \tag{3}$$

A solution y of (1) is *dominant* if $y_n \asymp P_n$ (so y_n/P_n tends to a nonzero constant), and the one-dimensional ray of *minimal* solutions m satisfies $m_n/y_n \to 0$, hence $m_n/P_n \to 0$. Writing $q = C\, y + D\, m$ with y normalized so that $y_n/P_n \to 1$,

$$C \;=\; R_\infty \;:=\; \lim_n Q_n \;=\; \lim_n \frac{q_n}{\prod_{k=1}^{n} b(k)}, \qquad \text{and} \qquad C \neq 0 \iff q \text{ is not minimal.} \tag{4}$$

This C is the object to decide. Equivalently, "q is exactly the minimal solution" (the degenerate seed) is exactly $C = 0$.

Remark 1. *The abstract dominant/minimal dichotomy and the uniqueness of the minimal ray are the content of Perron's theorem for $b(n) \to \infty$ and of Pincherle's theorem. [FLAG(operator): confirm Perron / Pincherle statement and locator] We invoke them only as context; §2–3 below are self-contained and elementary, and §4 is a numerical existence statement.*

2 The ratio criterion (dominant $\Rightarrow C \neq 0$)

Let $\rho_n = q_n/q_{n-1}$. From (1), wherever $\rho_{n-1} \neq 0$,

$$\rho_n = b(n) + \frac{1}{\rho_{n-1}}. \tag{5}$$

When $b(n) \to +\infty$ this Riccati form has two asymptotic branches: a dominant branch $\rho_n \sim b(n) \to +\infty$, and a minimal branch $\rho_n \sim -1/b(n+1) \to 0^-$.

Proposition 1 (Ratio criterion; **STRUCTURAL**, elementary)**.** *Suppose there is N_0 with $b(n) > 0$ for all $n \geq N_0$ and $q_{N_0-1}, q_{N_0} > 0$. Then $\rho_n > b(n)$ for every $n > N_0$, hence $\rho_n \to +\infty$, q is on the dominant branch, and $C \neq 0$.*

Proof. $\rho_{N_0} = q_{N_0}/q_{N_0-1} > 0$. If $\rho_{n-1} > 0$ then by (5) $\rho_n = b(n) + 1/\rho_{n-1} > b(n) > 0$, and the induction continues. Thus $\rho_n > b(n)$ for all $n > N_0$, and $b(n) \to +\infty$ gives $\rho_n \to +\infty$. A solution with $q_n/q_{n-1} \to +\infty$ cannot be minimal (the minimal ratio tends to 0), so q is dominant and $C \neq 0$ by (4). □

No external asymptotic theorem is used: the bound $\rho_n > b(n)$ is a one-line induction. (The determinant identity $\det M_n = -1$ only shows the propagation is a bijection; it does *not* by itself preserve "off the minimal ray," and is *not* used here.) Numerically, the dominant seed has $q_N/q_{N-1} = b(N)$ to machine precision, while the backward-computed minimal solution has $\rho_n^{\min} b(n+1) \to -1$ (`review_check_Cnonzero.py`).

3 Positivity-persistence: a sufficient condition with a machine-checked core

We now give the sufficient condition in the scaled frame (3), where its heart is a monotone–bounded argument that we have formalized.

Lemma 1 (Positivity persistence; engine **PROVEN**). *Suppose there is an index N_0 with* (i) $u_n > 0$ *for all $n > N_0$ (guaranteed once $b(n) > 0$ for all $n \geq N_0$), and* (ii) $Q_{N_0-1} \geq 0$ *and $Q_{N_0} > 0$. Then Q_n is positive and nondecreasing for $n \geq N_0$, the limit $R_\infty = \lim_n Q_n$ exists, and*

$$C = R_\infty \;\geq\; Q_{N_0} \;>\; 0.$$

In particular q is dominant and $C \neq 0$.

Proof. For $n > N_0$, $Q_n = Q_{n-1} + u_n Q_{n-2}$. By induction from (ii): if $Q_{n-2} \geq 0$ and $Q_{n-1} > 0$ then $Q_n = Q_{n-1} + u_n Q_{n-2} \geq Q_{n-1} > 0$, and (Q_{n-1}, Q_n) again satisfies the hypotheses. Hence $Q_n \geq Q_{n-1}$ for $n > N_0$, so the sequence is nondecreasing and bounded below by $Q_{N_0} > 0$. It is bounded above because $\sum_n u_n < \infty$ (as $u_n \sim \beta_d^{-2} n^{-2d}$ with $2d \geq 2$), giving $Q_n \leq Q_{N_0} \prod_{k>N_0}(1+u_k) < \infty$; a monotone bounded sequence converges, so $R_\infty \geq Q_{N_0} > 0$. □

What is machine-checked. The monotone and bounded-below steps are, in the canonical normalization $r_0 = r_1 = 1$, exactly the Lean lemmas `rseq_ge_one` and `rseq_mono` for the sequence $r_{n+2} = r_{n+1} + u_{n+2} r_n$ over any $u \geq 0$ (§6). These are parametric in the sequence u (hence in b), not in any numeric instance; their axiom cones are clean. Two honest scope caveats: (a) `rseq` fixes the canonical start $r_0 = r_1 = 1$, so Lean certifies exactly the $N_0 = 1$ instance; the shifted statement of Lemma 1 for $N_0 > 1$ is an elementary variant we have not formalized (grade **STRUCTURAL**); (b) the reduction "$b > 0 \Rightarrow u_n \geq 0$ and Q obeys (3)" is a one-line hand step, not inside Lean.

Corollary 1 (All-positive case; **PROVEN** engine + elementary reduction). *If $b(n) > 0$ for all $n \geq 1$, take $N_0 = 1$: then $Q_0 = Q_1 = 1 > 0$ and $u_n > 0$, so $C = R_\infty \geq 1$. In particular, for the family $b(n) = An^2 + Bn + C_0$ with integers $A, B, C_0 \geq 1$ (so $b(n) \geq A + B + C_0 \geq 3 > 0$ for all $n \geq 1$), $C \geq 1$ holds with no extra hypothesis.*

This recovers, and explains structurally, the strict positivity $\delta = \log R_\infty > 0$ that the companion quadratic growth-law deposit uses for the family $A, B, C_0 \geq 1$: there the constant δ is the continued-fraction correction beyond the elementary Γ-product, and Corollary 1 is precisely why it is positive.

Seed-index check (no $b(0)$ leak). A $\forall$-over-the-family claim must rule out an off-by-one at the seed, since $u_n = 1/(b(n-1)b(n))$ references $b(n-1)$. It does not leak: the scaled recurrence (3) begins at $n = 2$ with $u_2 = 1/(b(1)b(2))$, and $Q_0 = Q_1 = 1$ are pure seeds independent of $b(0)$. The smallest index of b used anywhere is $b(1)$; $b(0) = C_0$ never enters a denominator (mirrored in the Lean `rseq`, whose r_{n+2} references u_{n+2} from $n = 0$, i.e. u_2 upward, never u_0, u_1). This was checked at 50-digit precision for $(A, B, C_0) \in \{(1,1,1), (1,1,5), (3,1,1), (1,7,1), (2,5,9), (1,1,1000)\}$ including the boundary $A = B = C_0 = 1$: $b(0)$ untouched, the recurrence residual is $\sim 10^{-51}$, Q_n is nondecreasing and ≥ 1, and $C \in [1.00002, 1.066]$ (`check_seed_index.py`). **VERIFIED**.

4 Sharpness: the universal claim is false

The positivity hypothesis of Lemma 1 cannot be dropped. We disprove the universal statement by an explicit witness, following the discipline of trying to *break* the claim rather than confirm it.

Theorem 1 (Vanishing witness; **FALSE** for the universal claim, **VERIFIED**)**.** *The statement "$C \neq 0$ for every degree-d polynomial b with $\beta_d > 0$" is false. For*

$$b(n) = n^2 - 6.2803109563277\ldots\, n + 8.4205764270426\ldots = (n-\alpha^*)(n-\beta^*), \quad \alpha^* \approx 1.9401554781638748, \quad \beta^* = \alpha^* +$$

one has $C = R_\infty = 0$.

Numerical evidence (above the floor). At working precision 50 decimal digits (`verify_witness_precision.p`) refining the root by bisection drives R_∞ in lock-step with the bracket width, $R_\infty \propto (\alpha^* - \text{root})$ down to $\sim 10^{-40}$—the signature of a genuine *simple* zero, not a precision floor. At the refined α^*, $R_\infty = 3.756 \cdot 10^{-40}$ is stable across $N = 800, 1600, 3200, 6400$, while the dominant scale $\log_{10} \prod_{k \le N} |b(k)|$ grows $3934 \to 8846 \to 19634 \to 43140$; i.e. the seed is minimal by some 10^{44000} orders relative to the dominant solution. Transversality: $\alpha^* \pm 0.05$ give $R_\infty = +0.355$ and -4.32, order-one values of opposite sign. The vanishing is caused by the sign-varying transient—$b(1) = +3.140$, $b(2) = -0.140$, $b(3) = -1.420$, then $b(n) > 0$ for $n \geq 4$—so $b(1) > 0$ does *not* imply $C \neq 0$, and the cancellation lives in the seed/transient, exactly where Lemma 1's positivity hypothesis is needed.

Only ≥ 1 zero is claimed (poles, not a zero count). A coarse scan of $(\alpha, \beta) \mapsto R_\infty$ shows many sign changes, but the map has genuine *poles* wherever b has an integer root (some $b(k) = 0$, so $R_\infty = q_n/P_n \to \pm\infty$): for the gap-2.4 slice these occur at $\alpha \in \{1.6, 2.0, 2.6, 3.0\}$ (`check_bracket_continuity.py`). A sign-change count is therefore *not* a zero count, and we make no claim about the total number of zeros. The witness above is carried by a single bracket $[1.85, 1.98]$ that is *pole-free* (nearest integer collisions 1.6 and 2.0 lie outside), on which R_∞ is smooth and monotone, $+0.445 \to +0.002 \to -1.68$, a single clean crossing. The cancellation locus $\{b : C(b) = 0\}$ is thus nonempty; since $C(b)$ is a transcendental connection constant (not a polynomial in the coefficients), its zero set is a real-analytic variety of codimension one, here exhibited explicitly at one point.

5 The open point: $C = 0$ versus "entire"

Theorem 1 shows the dominant coefficient C vanishes for the witness. If one passes to the generating function $G(z) = \sum_n g_n z^n$ of the normalized denominator coefficients, a natural reading is "$C = 0 \Rightarrow G$ is entire." That implication is *not* established here. It rests on a dichotomy

$$\text{radius}(G) = R \iff \text{the singular local mode at } z = R \text{ is present} \iff C \neq 0,$$

whose nontrivial half requires that the holomorphic Frobenius modes of the ODE annihilating G have radius strictly greater than $R = d^d/\beta_d$ (for $d = 2$, $R = 4/A$, **VERIFIED** to ten digits in `review_check_Cnonzero.py`). That holds iff R is the *unique nearest* singular point of the annihilating ODE; if a second singular point sits at distance exactly R, a holomorphic mode could also have radius R and the dichotomy breaks. Establishing the unique-nearest-singularity premise (enumerate the singular points; prove R is the unique closest to the origin) is a separate, cleanly stated task, left as future work. Until it is closed, the honest reading of Theorem 1 is "the singular Frobenius coefficient vanishes," not "G is entire."

6 The Lean core

The accompanying Lean 4 project (`pcf-continuant`, Mathlib pinned via `lake-manifest.json`) formalizes the finitary structural core. The Casoratian identities are over an arbitrary commutative ring; the monotone/bounded-below lemmas are over $\mathbb{R}$. Verbatim signatures:

```
def IsSol (c s : N -> R) : Prop := forall n, s (n+2) = c (n+2) * s (n+1) + s n
def casoratian (y z : N -> R) (n : N) : R := y (n+1) * z n - y n * z (n+1)
theorem casoratian_step {c y z : N -> R} (hy : IsSol c y) (hz : IsSol c z) (n) :
    casoratian y z (n+1) = - casoratian y z n
theorem casoratian_eq   {c y z : N -> R} (hy : IsSol c y) (hz : IsSol c z) (n) :
    casoratian y z n = (-1)^n * casoratian y z 0
theorem pq_casoratian {b p q : N -> R}
    (hp : IsSol b p) (hq : IsSol b q)
    (hp0 : p 0 = 1) (hp1 : p 1 = b 1 + 1)
    (hq0 : q 0 = 1) (hq1 : q 1 = b 1) (n) :
    p (n+1) * q n - p n * q (n+1) = (-1)^n
theorem rseq_ge_one (u : N -> R) (hu : forall n, 0 <= u n) :
    forall n, 1 <= rseq u n /\ 1 <= rseq u (n+1)
theorem rseq_mono   (u : N -> R) (hu : forall n, 0 <= u n) (n) :
    rseq u n <= rseq u (n+1)
```

Running `lake env lean Check.lean` (`#print axioms`) yields the following cones, all subsets of {`propext`, `Classical.choice`, `Quot.sound`} with no `sorryAx`:

```
'PcfContinuant.casoratian_step' depends on axioms: [propext]
'PcfContinuant.casoratian_eq'   depends on axioms: [propext, Quot.sound]
'PcfContinuant.pq_casoratian'   depends on axioms: [propext, Quot.sound]
'PcfContinuant.rseq_ge_one'     depends on axioms: [propext, Classical.choice, Quot.sound]
'PcfContinuant.rseq_mono'       depends on axioms: [propext, Classical.choice, Quot.sound]
```

Hence the engine of Lemma 1 (Corollary 1 at $N_0 = 1$) is **PROVEN** in the strict sense used here.

7 Summary of claims and grades

Statement	Grade
$C = R_\infty = \lim q_n / \prod b$; $C \neq 0 \Leftrightarrow q$ not minimal	derived (§1); Casoratian piece PROVEN
Ratio criterion $\rho_n \to +\infty \Rightarrow C \neq 0$ (positive tail)	STRUCTURAL, elementary
Lemma 1 monotone/bounded engine ($N_0 = 1$)	PROVEN (Lean, clean cone)
Shifted Lemma 1 for $N_0 > 1$	STRUCTURAL
$b(n) > 0\ \forall n \Rightarrow C \geq 1$; deposited family $A, B, C_0 \geq 1$	engine PROVEN + elementary reduction
No $b(0)$ seed-index leak for deposited family	VERIFIED (dps 50)
Universal "$C \neq 0$ for all $\beta_d > 0$"	**FALSE** (Thm 1)
≥ 1 explicit zero ($C = 0$ witness)	VERIFIED (dps 50, $\sim 10^{-40}$)
Number of zeros / locus structure	open (poles $\neq$ zeros)
$C = 0 \Rightarrow G$ entire	CONDITIONAL on §5 (open)
Perron / Pincherle / Frobenius radius bound	context only; [FLAG(operator): unverified]

Reproducibility. All numerics are `mpmath` at the stated precision: `review_check_Cnonzero.py` (ratio criterion, $R = d^d/\beta_d$), `falsify_Czero.py` and `locate_Czero.py` (the witness), `verify_witness_precision.` (simple-zero/precision check), `check_bracket_continuity.py` (pole audit), `check_seed_index.py`

(no $b(0)$ leak). The Lean core builds against the pinned Mathlib in `lake-manifest.json`; `Check.lean` reproduces the axiom cones above.

Acknowledgement of method. This note is the product of an adversarial review cycle in which each load-bearing claim was attacked numerically before being asserted. That process retracted an earlier overcount ("many zeros by the intermediate value theorem," invalid because of the poles), corrected the value of R, demoted a determinant argument that did not prove what it appeared to, and upgraded the deposited-family positivity to a machine-checked statement. The two remaining open items (§5; the literature locators) are stated as such rather than papered over.

Code and data availability. The numerical witnesses `review_check_Cnonzero.py`, `review_check_transient.p` `falsify_Czero.py`, `locate_Czero.py`, `verify_witness_precision.py`, `check_bracket_continuity.py`, and `check_seed_index.py`, together with the Lean core `lean/pcf_continuant/` (`Check.lean` reproduces the axiom cones of §6), are available in the repository `papanokechi/pcf-continuant` (`https://github.com/papanokechi/pcf-continuant`).

AI-assistance disclosure. Prepared with AI assistance (GitHub Copilot; Anthropic Claude) for drafting, computer-algebra verification, and manuscript preparation. All Lean axiom cones were obtained by compiling against the pinned toolchain (`lake-manifest.json`); the numerical witnesses were run under `mpmath` at 50-digit precision. The author takes responsibility for the content and for the placement of each claim within the four-grade epistemic partition.